FIELD AND SERVICE ROBOTICS

Quadruped MPC, Rimless Wheel Dynamics and Underwater Robot Modeling

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GitHub repository

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1 The Buoyancy Effect

Buoyancy is a **hydrostatic effect** that arises when a body is submerged in a fluid under the influence of gravity. It manifests as an upward force exerted by the fluid, opposing the weight of the object. Unlike hydrodynamic forces (e.g., drag or lift), buoyancy is not a function of the relative velocity between the object and the fluid.
This force results from a pressure difference across the vertical extent of the submerged body. Specifically, in a fluid column, pressure increases with depth due to the weight of the overlying layers. As a result, the pressure exerted on the lower surface of a submerged object is higher than on the upper surface, generating a net upward force. Buoyancy opposes gravity and is directly proportional to the submerged volume of the body, the fluid density, and gravitational acceleration:

$$b = \rho \Delta \|\bar{\mathbf{g}}\|, \tag{1}$$

where ρ is the fluid density, Δ is the volume of displaced fluid, and $\bar{\mathbf{g}}$ is the gravity acceleration vector.
A critical aspect of buoyancy modeling involves recognizing that the buoyant force typically acts at the *center of buoyancy*, which corresponds to the centroid of the displaced fluid volume. This point generally does not coincide with the robot's *center of mass*. Their spatial separation induces a torque that can affect the robot's attitude and must be accounted for in the dynamic model.
In body-fixed frame, the total wrench due to gravity and buoyancy is:

$$\mathbf{g}_{rb}^{b} = -\begin{bmatrix} \mathbf{f}_{g}^{b} + \mathbf{f}_{b}^{b} \\ S(\mathbf{r}_{c}^{b})\mathbf{f}_{g}^{b} + S(\mathbf{r}_{b}^{b})\mathbf{f}_{b}^{b} \end{bmatrix}, \tag{2}$$

where $S(\cdot)$ denotes the skew-symmetric matrix operator, $\mathbf{r}_{c}^{b}$ and $\mathbf{r}_{b}^{b}$ are the position vectors of the center of mass and center of buoyancy, respectively, expressed in the body frame, and:

$$\mathbf{f}_{b}^{b} = -R_{b}^{T}\begin{bmatrix} 0 \\ 0 \\ b \end{bmatrix} \tag{3}$$

is the buoyancy force acts at the center of buoyancy.
Even a small offset between the center of mass and the center of buoyancy can generate a non-negligible torque. This torque must be compensated by the control system, especially in large underwater vehicles, where the required correction may demand significant thrust. Such torques play an important role in stability analysis and must be addressed during control design to ensure reliable attitude regulation. In contrast, buoyancy is typically neglected in aerial robotics. This simplification is justified by the vast difference in density between air and typical aerial robots, which are often composed of dense materials such as carbon fiber, metals, or composite alloys. As a result, the buoyant force in air is several orders of magnitude smaller than the gravitational force and has a negligible effect on the dynamics. For this reason, it is possible to exclude buoyancy from aerial dynamic models, thereby simplifying the control design without compromising model accuracy.

2 Assessment of Modeling Principles in Underwater Robotics

(a) False. The added mass effect does not correspond to a physical load or an actual increase in structural mass. Instead, it reflects the reaction forces exerted by the fluid as it is accelerated by the motion of the robot. These forces are modeled by an added mass matrix, which captures the apparent inertia due to the displaced fluid. This matrix may not be positive definite and should not be interpreted as a conventional mass term added to the robot's body.

(b) True. The added mass effect is particularly relevant in underwater robotics because the density of water is comparable to that of the robot itself. This leads to significant inertia being transferred from the fluid to the robot dynamics. In contrast, in aerial robotics, the much lower density of air makes the added mass negligible. Hence, in underwater environments, neglecting this effect would result in inaccurate modeling.

(c) True. The damping effect originates from the viscous nature of the fluid and is typically modeled as a combination of drag and lift forces acting at the center of mass of the body. These forces oppose the relative motion between the robot and the fluid and are dissipative in nature. As such, they play a beneficial role in the stability analysis of underwater systems by introducing passive damping into the dynamics.

(d) False. The ocean current is typically modeled as a constant and irrotational velocity field, meaning it does not vary with time or space within the region of interest. This assumption simplifies the modeling of environmental disturbances acting on underwater robots. However, a critical aspect is the choice of reference frame: although the current may be constant in the world frame, it would appear time-varying if expressed in the body-fixed frame, due to the robot's own motion. For this reason, the ocean current is always defined in the world frame and then transformed into the body frame when needed.
In this context, the ocean current vector is typically assumed to have translational components only, with zero angular velocity. Its structure and time derivative are defined as:

$$\boldsymbol{\nu}_c = \begin{bmatrix} v_{c,x} \\ v_{c,y} \\ v_{c,z} \\ 0 \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^6, \quad \dot{\boldsymbol{\nu}}_c = \mathbf{0}_6. \tag{4}$$

This reflects the assumption of a steady and uniform ocean flow acting on the robot throughout the mission.
To properly account for the influence of the current in the robot's dynamics, it is incorporated through the relative velocity between the robot and the fluid. This relative velocity, expressed in the body-fixed frame, is given by:

$$\boldsymbol{\nu}_r = \begin{bmatrix} \dot{\boldsymbol{p}}_b^b \\ \boldsymbol{\omega}_b^b \end{bmatrix} - R_b^\top \boldsymbol{\nu}_c, \tag{5}$$

This formulation is fundamental for the correct computation of Coriolis, centripetal and damping effects.

3 Quadruped Simulations and Analysis of Results

3.1 Evaluation with Default Parameters and Mass Variations

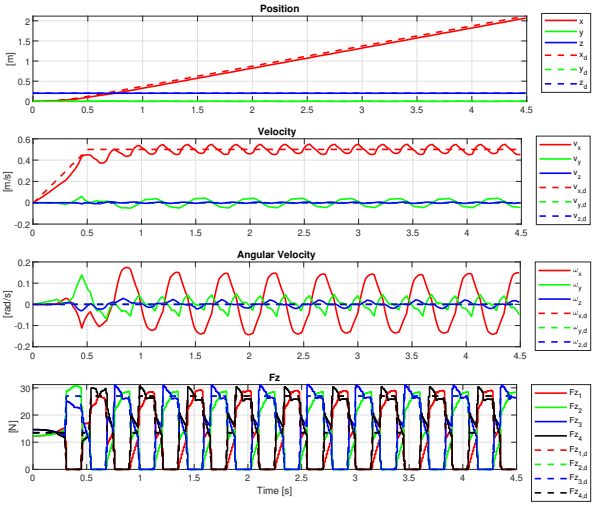
I have slightly modified the `fig_animate.m` file by adding code at lines 5-6, 30, and 133-157, 177-229 to enable the automatic saving of the following plots in EPS format. This eliminates the need for manual screenshots at the end of each simulation and significantly improves the quality and consistency of the generated figures. **A first analysis** was carried out on a set of simulations using the default parameters:

$$gait \in \{0, \ldots, 5\}, \quad m = 5.5 \text{ kg}, \quad v_{x,\mathrm{d}} = 0.5 \text{ m/s}, \quad v_{y,\mathrm{d}} = 0 \text{ m/s}, \quad \mu = 1.$$

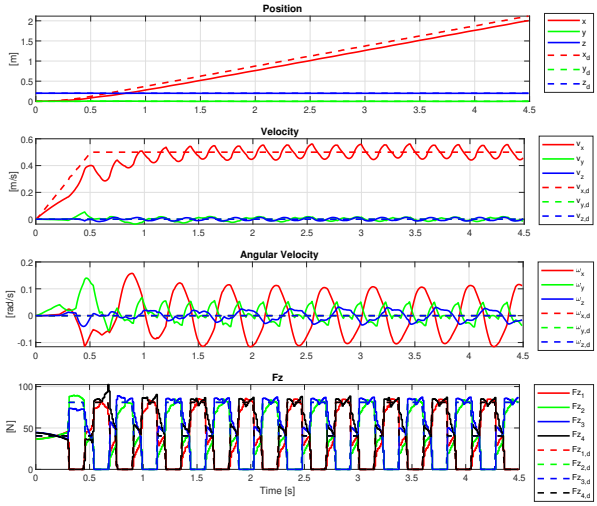
I analyzed the behavior of the quadruped robot for the following gait patterns: trot, bound, pacing, gallop, trot-run, and crawl. For each gait, I considered the tracking of the position and desired velocity, the smoothness of the ground reaction forces (F_z), the body angular velocity, and the overall stability of the motion. The results are summarized below. To assess the influence of mass on gaits performance, two simulations were conducted with the same gait pattern, velocity reference and friction coefficient, but **different total mass values**: 16.5 kg and 2.75 kg, obtained by tripling and halving the original mass (5.5 kg). For vision clarity **the other tests** with the variation of friction coefficient and desired velocity will be presented in the next sections. **All videos can be viewed on Github in the appropriate folders.**

3.1.1 Trot gait

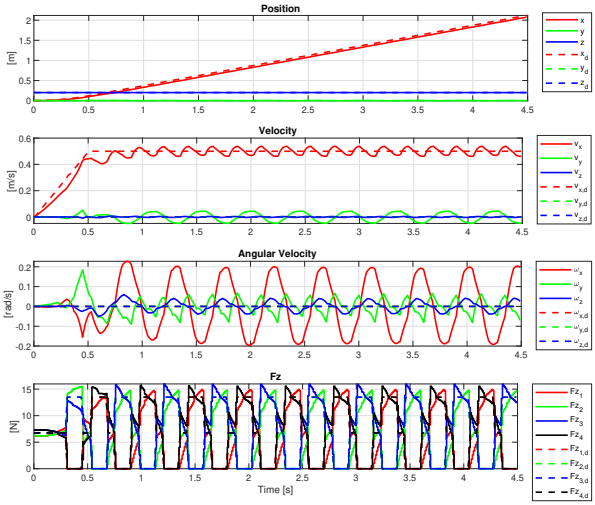
The trot gait is among the most stable and efficient patterns for quadrupedal locomotion. The baseline simulation (Figure 1a) shows symmetric and periodic locomotion, with a steady forward velocity of approximately 0.46 m/s. Ground reaction forces (F_z) are evenly distributed and reveal alternating diagonal contacts. Angular velocity remains bounded, and the body motion is smooth and well-coordinated.
With $m = 16.5$ kg (tripled), the robot exhibits damped and smooth body oscillations. Vertical forces are stable, and pitch/roll remain small. Tracking is slightly slower, but postural stability improves. With $m = 2.75$ kg (halved), the robot becomes more reactive but also more unstable. Angular oscillations increase, and GRFs become less regular. The CoM is more sensitive to impacts, although velocity tracking remains effective.



(a) Default: $m = 5.5\,\text{kg}$.



(b) Heavy: $m = 16.5\,\text{kg}$.



(c) Light: $m = 2.75\,\text{kg}$.

Figure 1: Trot gait under different mass conditions.

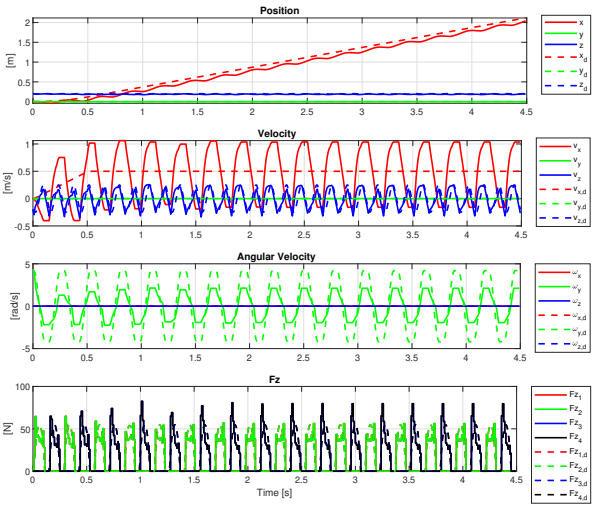
3.1.2 Bound gait

The bound gait (Figures 2a, 2b, 2c) is characterized by coordinated front-hind contact patterns. In the default configuration with $m = 5.5\,\text{kg}$, the robot follows the reference velocity of $0.5\,\text{m/s}$ with regular oscillations, though peak values exceed the reference. The linear and angular velocities exhibit stable, periodic fluctuations. The ground reaction forces (GRFs) display alternating and symmetric patterns, confirming proper lift and flight phases typical of a well-executed bounding gait.

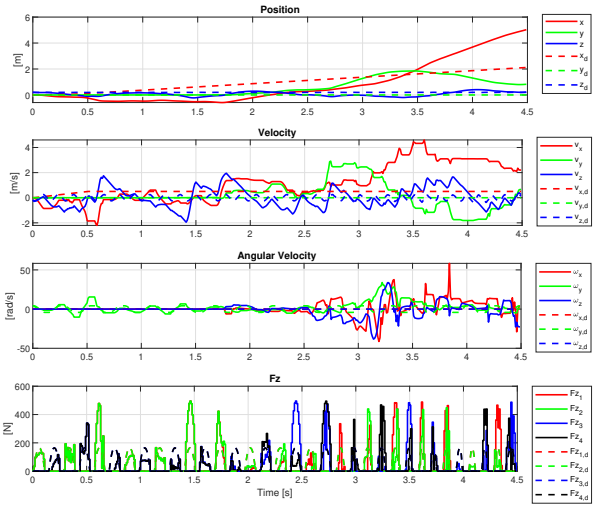
When the mass is increased to $m = 16.5\,\text{kg}$, the robot exhibits erratic motion with poor coordination. Despite the same reference velocity, the position and velocity plots show inconsistent tracking and instability. The angular velocity becomes noisy, and the GRFs display high-magnitude fluctuations and loss of symmetry, suggesting loss of regular gait and degraded performance.

Conversely, the lightweight case with $m = 2.75\,\text{kg}$ yields improved agility. The position and velocity follow the reference trajectory more accurately, with frequent and well-defined lift phases. Angular oscillations remain regular and symmetric, while GRFs are

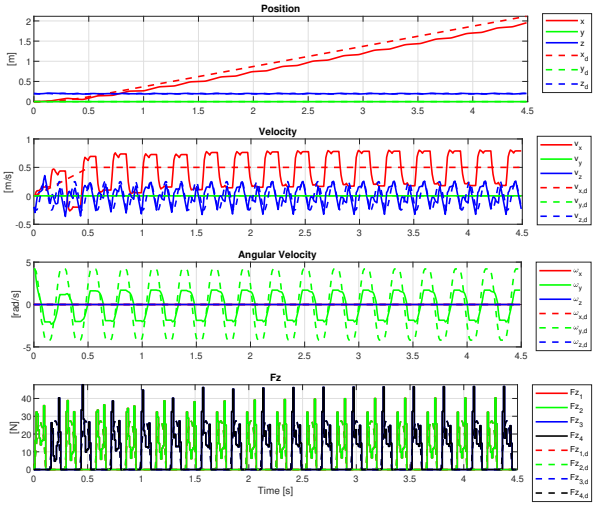
smooth and periodic, confirming dynamic stability and successful bounding.



(a) Default: $m = 5.5\,\text{kg}$.



(b) Heavy: $m = 16.5\,\text{kg}$.

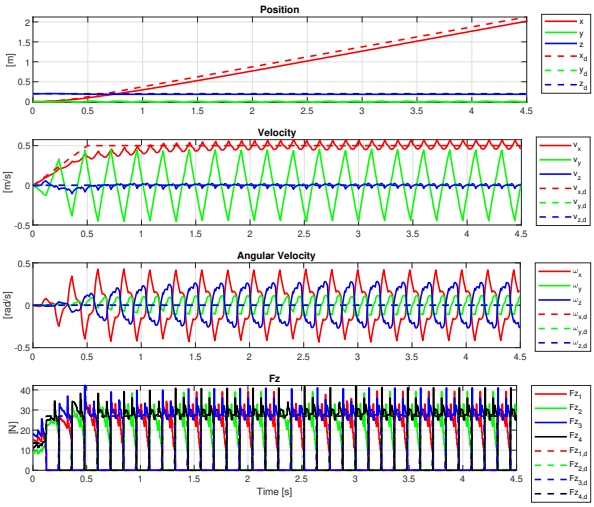


(c) Light: $m = 2.75\,\text{kg}$.

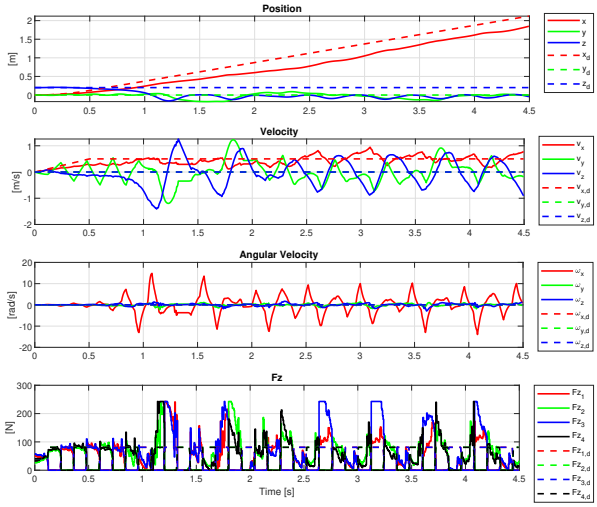
Figure 2: Bound gait under different mass conditions.

3.1.3 Pacing gait

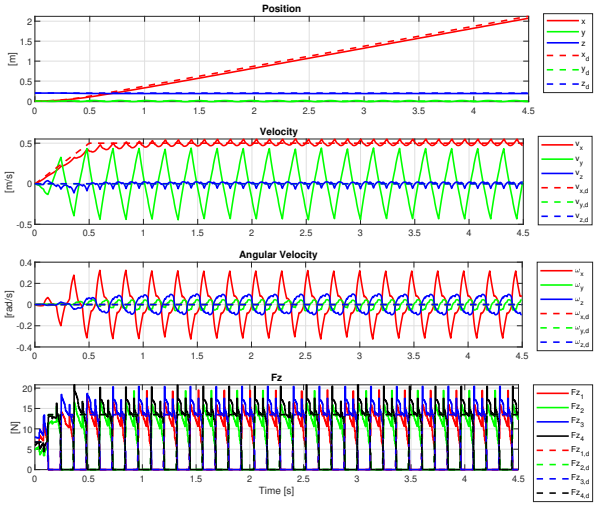
The pacing gait (Figures 3a, 3b, 3c) uses lateral leg pairs in phase.
With default mass, the robot reaches $v_x \approx 0.52$,m/s, but experiences lateral oscillations and GRFs with reduced symmetry compared to trot.
The heavier mass inhibits gait initiation: no motion occurs because the simulation failed.
The lighter robot performs effective pacing with stable leg coordination and smooth forward progression. Tracking remains accurate and energetic.



(a) Default: $m = 5.5\,\text{kg}$.



(b) Heavy: $m = 16.5\,\text{kg}$.



(c) Light: $m = 2.75\,\text{kg}$.

Figure 3: Pacing gait under different mass conditions.

3.1.4 Gallop Gait

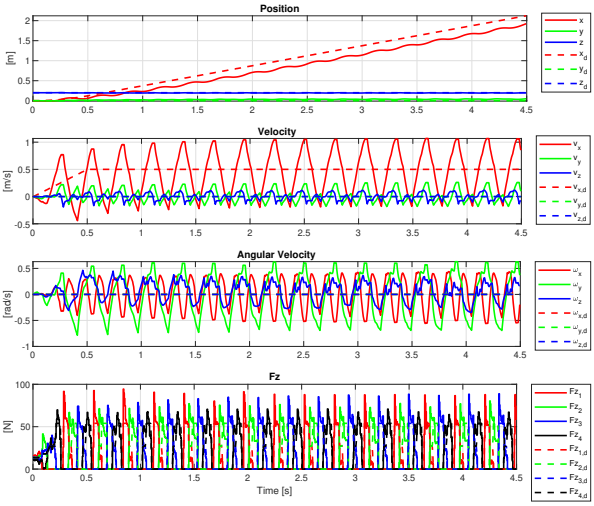
The **gallop** is a fast, asymmetric gait involving alternating aerial and stance phases, typically used for high-speed motion.

In the standard configuration (Figure 4a), the robot accelerates beyond the reference velocity ($v_d = 0.5\,\text{m/s}$), reaching nearly $v = 1.1\,\text{m/s}$. The gait features strong ground impacts and irregular GRFs, indicating impulsive dynamics and limited damping. Angular velocities present high variability, especially in pitch and roll, revealing limited postural stability. The CoM oscillates significantly, following a jagged path, typical of an underdamped high-energy gait.

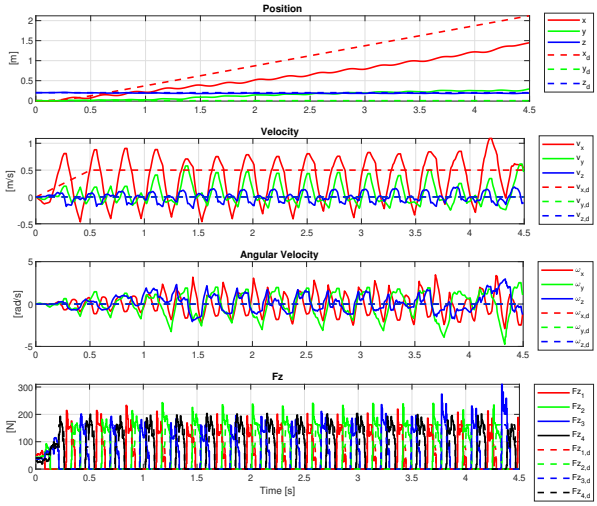
With increased mass (Figure 4b), the robot's motion is visibly restrained. The build-up of forward velocity is slow, and aerial phases are largely suppressed. Although the gait maintains symmetry and smooth angular dynamics, the system lacks agility and fails to exploit the full galloping pattern. The CoM remains close to the ground and exhibits minimal vertical motion.

The lightweight configuration (Figure 4c) enables a vivid galloping pattern. The robot

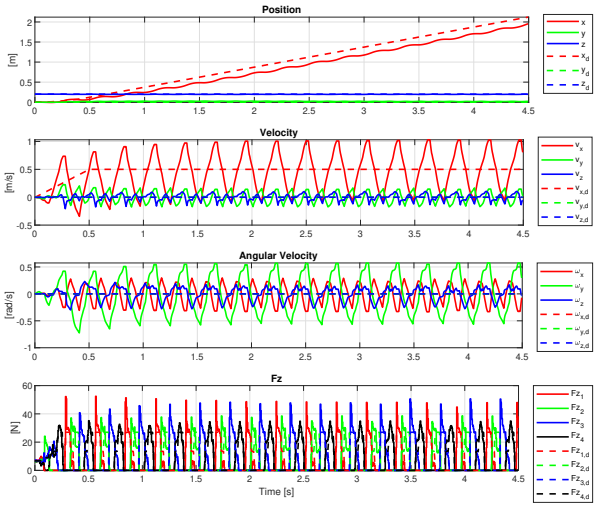
shows fast leg swings, distinct aerial phases, and dynamic alternation of contacts. The CoM follows a rhythmic oscillatory trajectory with significant vertical excursion, and the robot rapidly reaches the desired velocity. However, angular velocity increases, and GRFs are more spiked, indicating harsher impacts and reduced damping. The motion is agile and expressive but demands more from the controller to preserve balance.
In conclusion, heavier mass improves gallop smoothness but limits speed and agility, while lighter mass enhances dynamism and flight phases at the cost of control effort and postural stability.



(a) Default: $m = 5.5\,\text{kg}$.



(b) Heavy: $m = 16.5\,\text{kg}$.



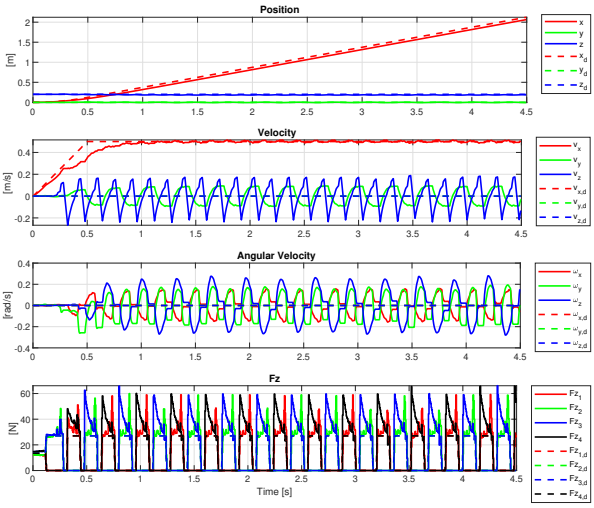
(c) Light: $m = 2.75\,\text{kg}$.

Figure 4: Gallop gait under different mass conditions.

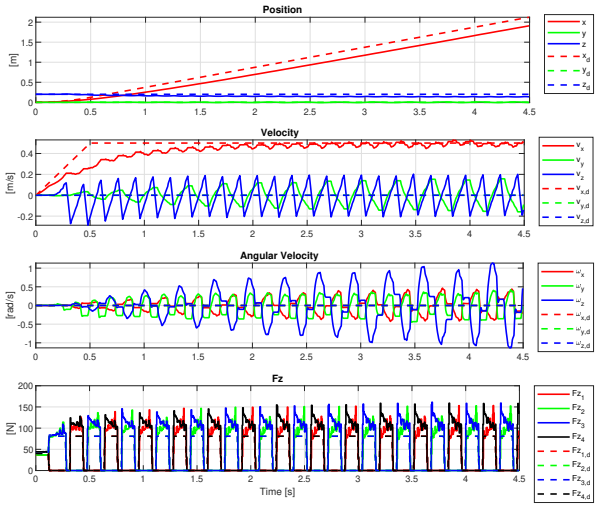
3.1.5 Trot-Run Gait

With standard parameters, the gait achieves excellent velocity tracking ($v \approx 0.49\,\text{m/s}$), and exhibits smooth GRFs with periodic and symmetric profiles. Angular velocity components remain bounded. Compared to trot, this gait introduces brief flight phases, which improve CoM dynamism while preserving overall stability.
As shown in Figure 5b, the target velocity is reached only partially, and the response to

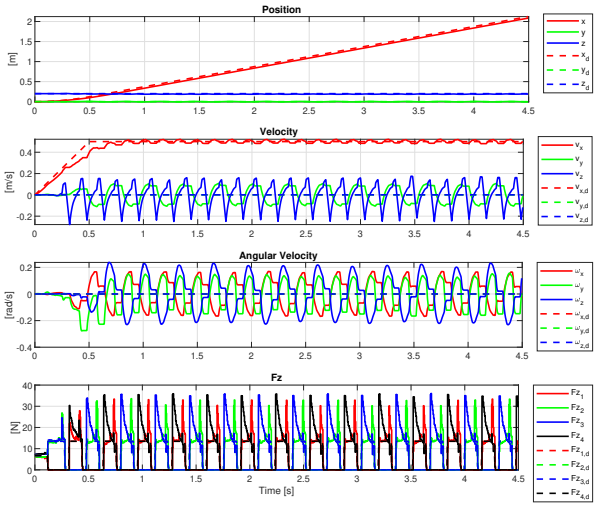
control inputs is less prompt. About the GRFs, in this case the magnitude is larger than the default case. However, the CoM progresses more slowly and slightly lags behind the reference.
In contrast, the light configuration (Figure 5c) enables fast convergence to the target velocity and more accurate CoM tracking, with minimal positional error. However, the system becomes sensitive to impacts: F_z peaks rise sharply, and angular velocity oscillations increase. The motion is more agile but less composed.



(a) Default: $m = 5.5\,\text{kg}$.



(b) Heavy: $m = 16.5\,\text{kg}$.



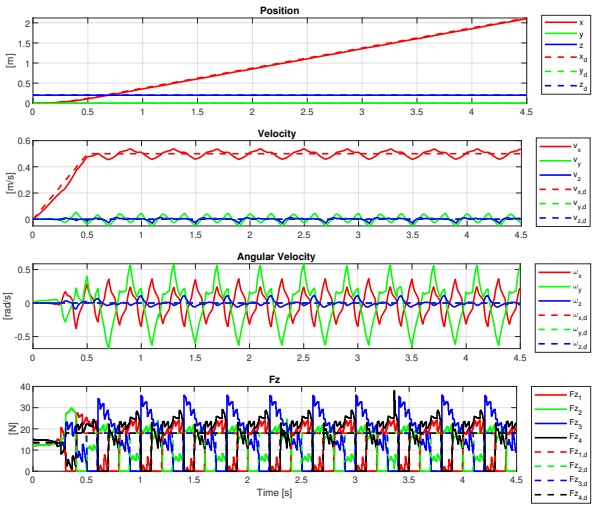
(c) Light: $m = 2.75\,\text{kg}$.

Figure 5: Trot-run gait under different mass conditions.

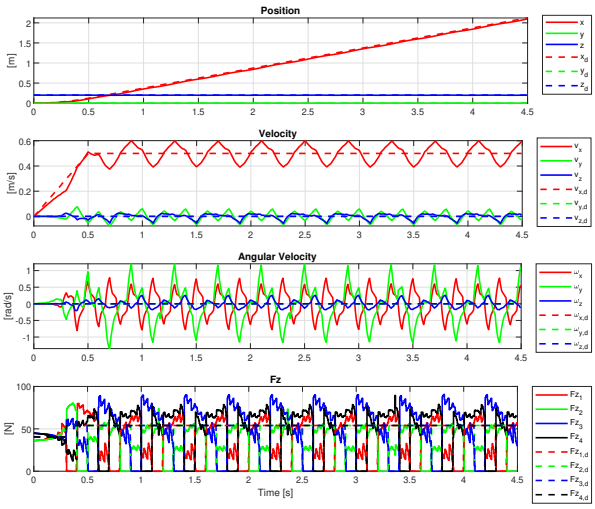
3.1.6 Crawl Gait

The **crawl** gait is a quasi-static, highly stable locomotion pattern involving sequential leg motion and continuous ground contact.
The robot tracks the reference velocity well ($v \approx 0.52\,\text{m/s}$), maintaining three legs in contact at all times. GRFs are low in magnitude but frequent, and angular velocities are limited. However, some asymmetries emerge due to sequential foot placement, making the gait more mechanically demanding.

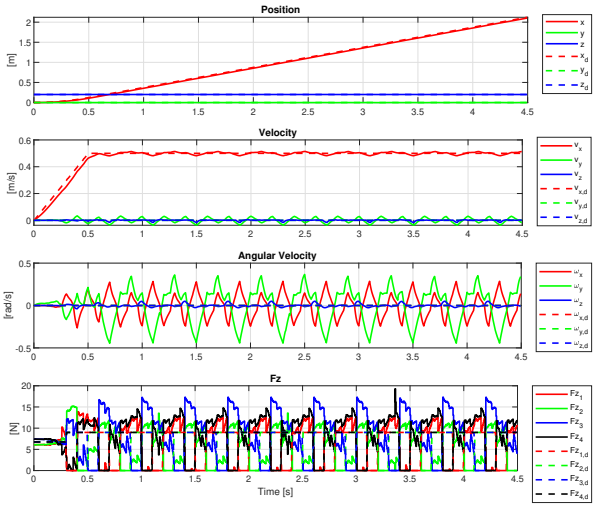
With a heavier mass (Figure 6b), GRFs rise to 55–60 N and appear squared due to long stance durations. Body oscillations are minimal, and CoM tracking is satisfactory but convergence to desired velocity is slow due to mass-induced inertia.
The light robot (Figure 6c) shows remarkable agility, rapidly reaching the target velocity and achieving sub-centimeter CoM error. GRFs are low and sawtooth-shaped, reflecting quick foot transitions.



(a) Default: $m = 5.5\,\text{kg}$.



(b) Heavy: $m = 16.5\,\text{kg}$.



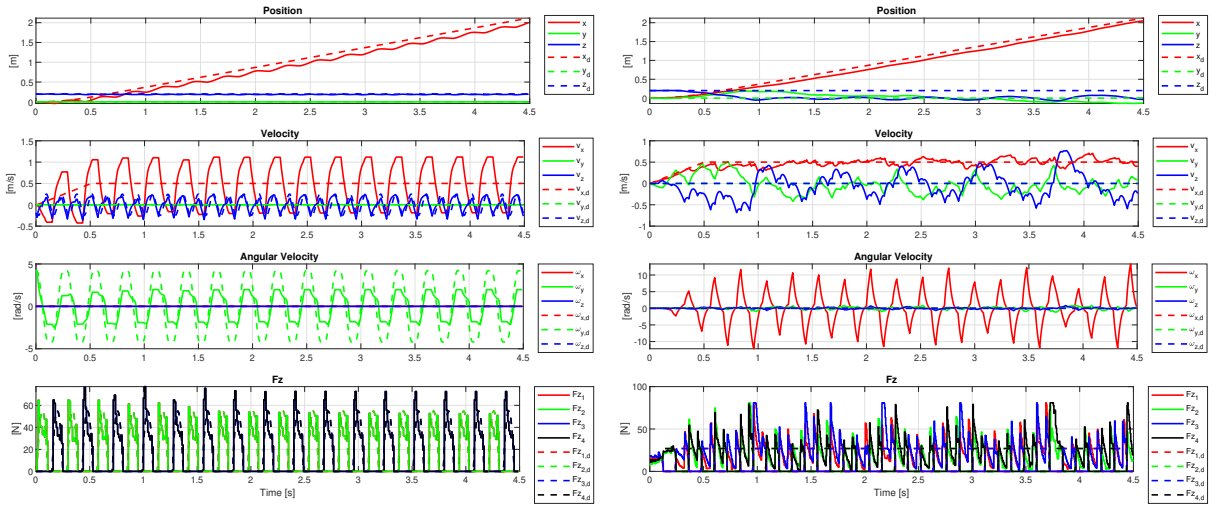
(c) Light: $m = 2.75\,\text{kg}$.

Figure 6: Crawl gait under different mass conditions.

3.2 Evaluation under Friction Variations

To assess the influence of friction on locomotion quality, the coefficient μ was varied from its nominal value ($\mu = 1$) to two alternative configurations: $\mu = 0.4$ and $\mu = 1.2$, the other parameters are the default ones. Each gait was tested under both modified values. The results indicate that increasing friction to $\mu = 1.2$ does not produce substantial changes in most gaits compared to the default case. Probably, the most notable negative effects were evident in the bound gait, as it's possible to see in (Figure 7a), the higher friction value degraded performance. The tracking of the position is slightly worse than

the default case. The GRFs present peaks with higher magnitude. This suggests that the benefits of increased friction depend on the specific gait dynamics and the role of aerial phases in the motion strategy.



(a) Bound gait with $\mu = 1.2$.

(b) Pacing gait with $\mu = 0.4$.

Figure 7: Effect of friction variation on bound and pacing gaits.

When the friction coefficient is decreased to $\mu = 0.4$, no significant deviations are observed in the behavior of the *trot*, *trot-run*, and *crawl* gaits compared to the standard configuration. These gaits maintain consistent reference tracking and postural stability despite the reduced friction.

In contrast, *bound*, *pacing*, and *gallop* gaits exhibit noticeable performance degradation. The reduced traction leads to impaired tracking of the velocity reference and a general decline in balance and coordination. In particular, the pacing gait it's the one which becomes more unstable under low-friction conditions, as illustrated in Figure 7b, where the robot struggles to maintain consistent contact and posture.

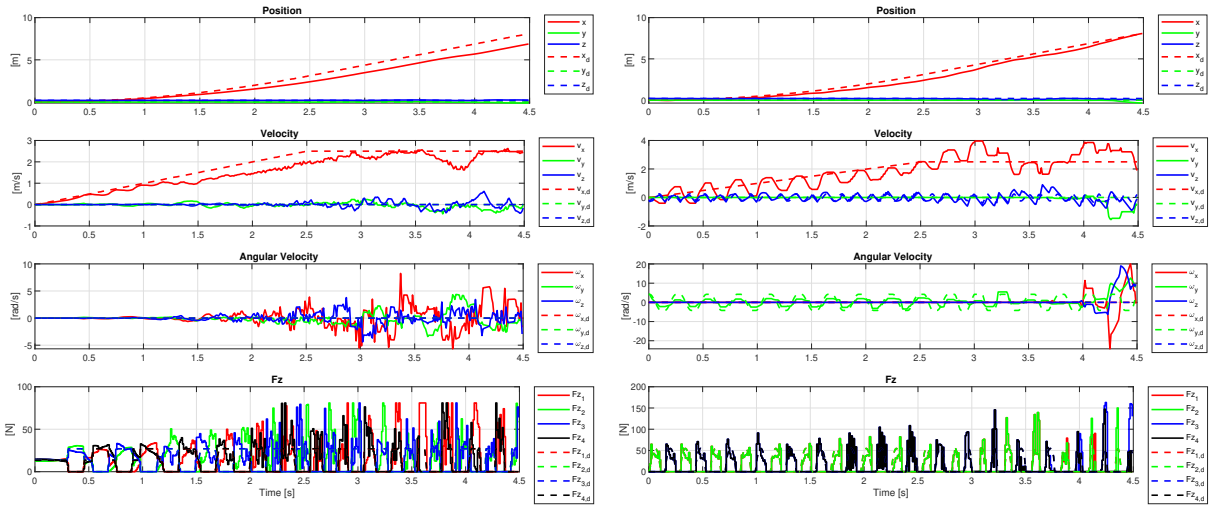
3.3 Evaluation under Increased Velocity

To evaluate gait performance under higher-speed locomotion, all gait patterns were tested with a desired forward velocity of $v_d = 2.5\,\text{m/s}$. The outcomes are reported in Figures 8 to 10, with a focus on tracking accuracy, angular stability, ground reaction forces F_z, and postural balance.

The **trot gait** (Figure 8a) achieves nearly complete velocity tracking, reaching about 2.45 m/s, close to the desired reference. However, this performance comes with a cost: angular velocity components exhibit large oscillations, reaching amplitudes up to $\pm 5\,\text{rad/s}$, which compromises the overall postural stability. The center of mass (CoM) follows the desired trajectory with acceptable accuracy, although not perfectly aligned. Ground reaction forces F_z remain periodic and coordinated across legs, but their magnitudes grow significantly to support the increased dynamics. Overall, the trot gait remains almost completely effective under high-speed conditions, but with evident degradation in angular stability.

In contrast, the **bound gait** (Figure 8b) showed signs of instability. Although the robot managed to accelerate near the target velocity, the motion became irregular with strong

impacts. The F_z profiles revealed large peaks, and the angular velocity curves indicated severe pitch oscillations, leading to partial loss of balance.

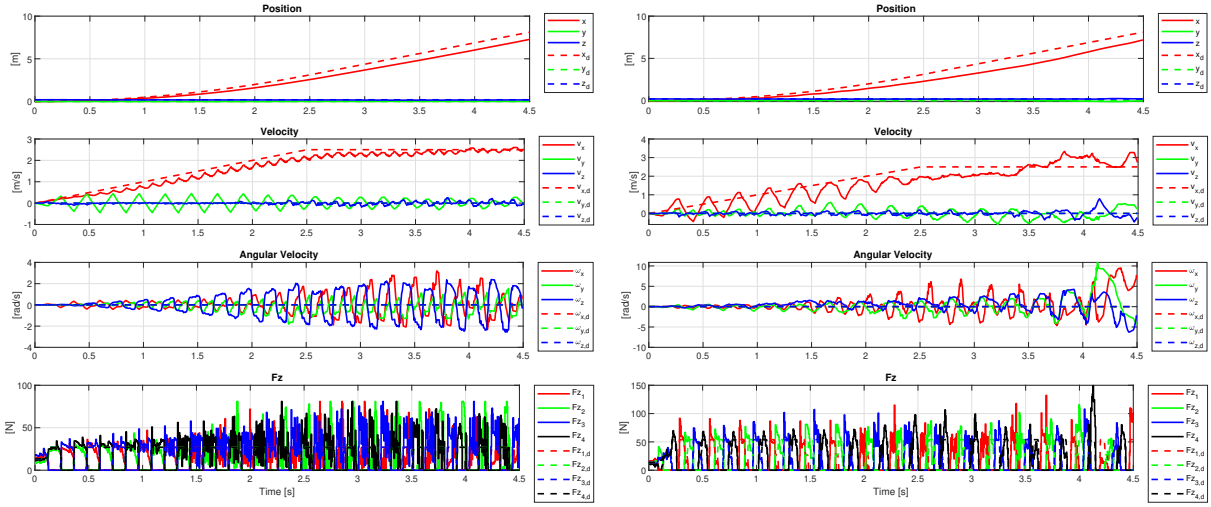


(a) Trot gait at $v_d = 2.5\,\mathrm{m/s}$.

(b) Bound gait at $v_d = 2.5\,\mathrm{m/s}$.

Figure 8: Performance comparison of trot and bound gaits at high speed.

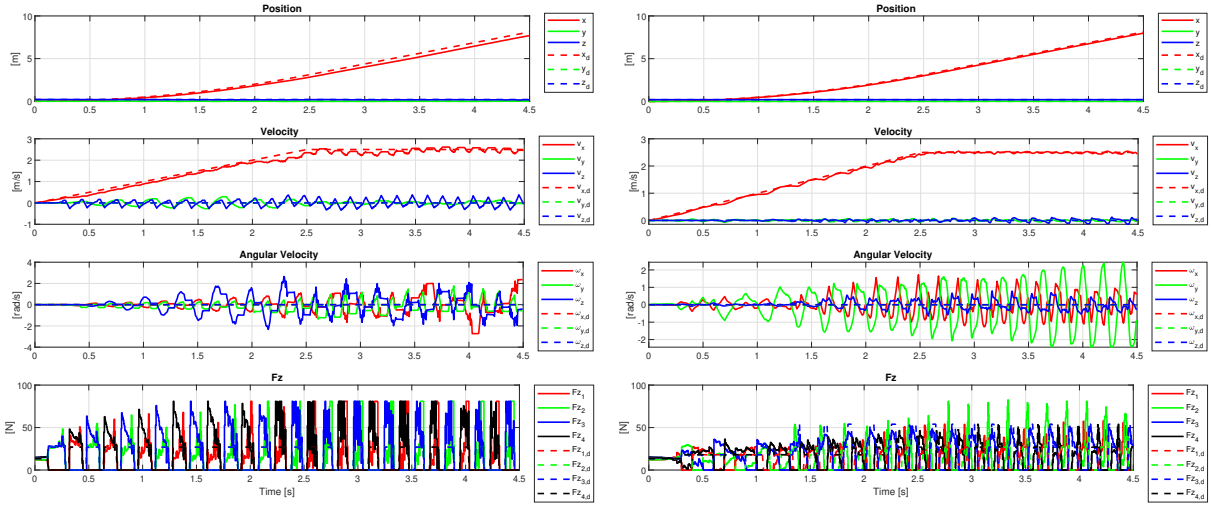
The **pacing gait** (Figure 9a) struggled at this speed. Velocity tracking was poor, both linear and angular components exhibited high-frequency oscillations. The F_z signals were erratic and asymmetric, reflecting uncoordinated leg activity and poor gait stability.
In the **gallop gait**, (Figure 9b), the robot fails to maintain a stable posture during galloping. Although the target velocity of 2.5 m/s is approximately reached, the body exhibits a full 360-degree rotation during motion, indicating a dramatic loss of postural control. This behavior reveals a breakdown in the regulation of pitch and roll dynamics, as the system is unable to counteract the impulsive vertical forces generated at foot contacts. The ground reaction forces F_z appear highly irregular and discontinuous, highlighting poor coordination and unbalanced impacts. Despite continuous forward progression of the CoM, the gait becomes chaotic and dynamically unstable, rendering galloping ineffective at this velocity without enhanced stabilization strategies.



(a) Pacing gait at $v_d = 2.5\,\mathrm{m/s}$. (b) Gallop gait at $v_d = 2.5\,\mathrm{m/s}$.

Figure 9: Performance comparison of pacing and gallop gaits at high speed.

In the **trot-run gait** (Figure 10a), the robot successfully tracked the velocity reference, combining rapid progression with moderate aerial dynamics. The CoM followed the reference trajectory closely, and F_z signals showed regular alternation. Angular oscillations were more pronounced than in the trot case but remained bounded and periodic.
Finally, the **crawl gait** (Figure 10b) adapts well to the increased target velocity. Although primarily designed for low-speed locomotion, the robot manages to track the reference trajectory with minimal error. The center of mass velocity stabilizes around $2.5\,\mathrm{m/s}$, ground reaction forces F_z remain structured and periodic, and angular velocities are well regulated. The system exhibits a highly controlled stepping pattern, though the high duty cycle and absence of flight phases render the motion less efficient at this speed.



(a) Trot-run gait at $v_d = 2.5\,\mathrm{m/s}$. (b) Crawl gait at $v_d = 2.5\,\mathrm{m/s}$.

Figure 10: Performance comparison of trot-run and crawl gaits at high speed.

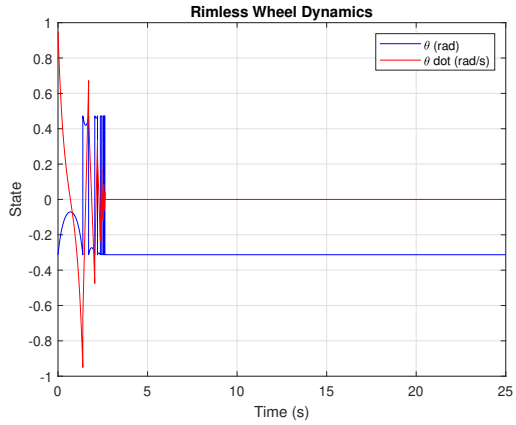
4 Rimless Wheel Dynamics

4.1 Different Initial Angular Velocities

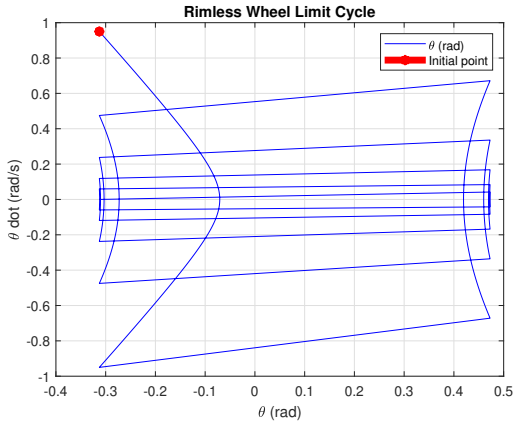
To investigate the stability and convergence behavior of the rimless wheel model, I ran three simulations by varying the initial angular velocity $\dot{\theta}_0$ while keeping the system parameters fixed: $\dot{\theta}_0 = 0.95$, -0.095, and 1.8 rad/s. The goal was to identify possible equilibria or limit cycles and their respective basins of attraction.
For the nominal value $\dot{\theta}_0 = 0.95$ rad/s, the system initially shows transient oscillations but ultimately fails to sustain motion, coming to rest after a few impacts. Both the phase portrait (Fig. 11b) and the time-domain plot (Fig. 11a) indicate that the angular velocity decreases to zero and the system stabilizes at a static equilibrium. This behavior is consistent with theoretical expectations: the initial velocity is slightly below the critical threshold

$$\omega_1 = \sqrt{2 \cdot \frac{g}{l}(1 - \cos(\gamma - \alpha))} \approx 0.9754 \text{ rad/s},$$

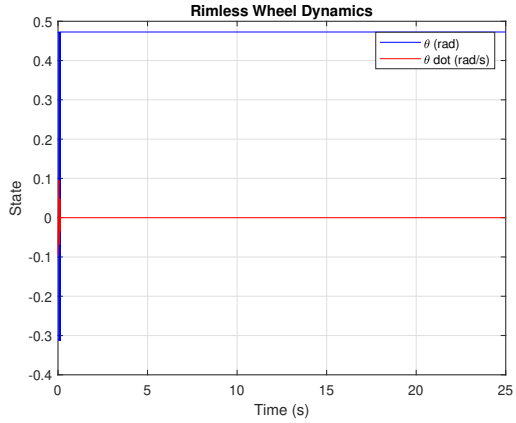
which represents the minimum angular velocity required to overcome the gravitational potential difference and initiate walking motion after impact.
With a lower energy level, i.e., $\dot{\theta}_0 = -0.095$ rad/s, the system exhibits almost no motion and rapidly settles into a resting state. The wheel lacks the kinetic energy required to transition into a walking gait. This behavior is evident from the decaying trajectory in the phase plot (Fig. 11d) and the minimal angular displacement in the time evolution (Fig. 11c).
Conversely, for a high initial angular velocity $\dot{\theta}_0 = 1.8$ rad/s, the system converges to a stable limit cycle. After an initial transient, the trajectory evolves toward a periodic orbit with larger angular excursions and sustained velocity. This can be observed clearly in the structured phase portrait (Fig. 11f) and the periodic dynamics (Fig. 11e).
Regarding the final equilibria, both $\dot{\theta}_0 = -0.095$ and 0.95 converge to the static equilibrium, whereas $\dot{\theta}_0 = 1.8$ leads to a dynamic steady-state walking pattern. Therefore, the system supports two distinct attractors: a static equilibrium and a dynamic limit cycle. The presence of multiple limit cycles was not observed in this analysis, though other cycles might exist for different initial conditions.
The basin of attraction of the static equilibrium includes low and moderate initial velocities insufficient to maintain periodic motion. In contrast, sufficiently high initial energy, as in the case of $\dot{\theta}_0 = 1.8$, places the system within the basin of attraction of the walking limit cycle.
Overall, the system presents two distinct attractors: a static equilibrium and a dynamic walking limit cycle. The equilibrium state is reached for insufficient initial angular velocities, while the dynamic limit cycle emerges when the initial energy exceeds the threshold ω_1.



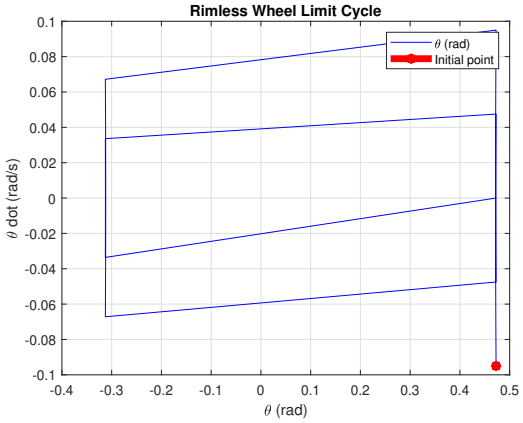
(a) State evolution for $\dot{\theta}_0 = 0.95$ rad/s.



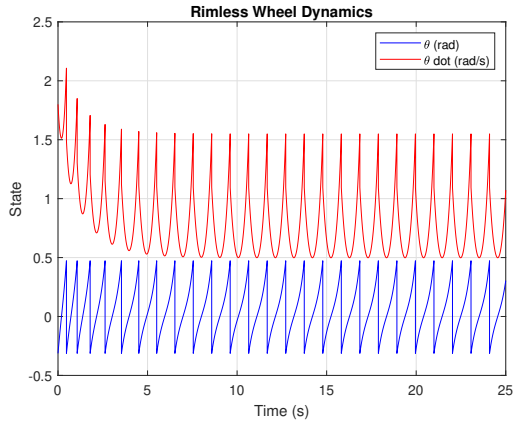
(b) Limit cycle for $\dot{\theta}_0 = 0.95$ rad/s.



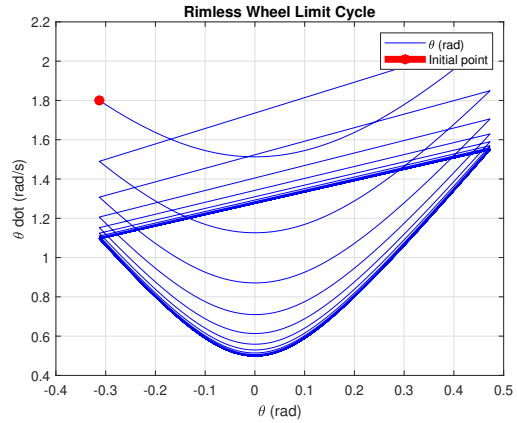
(c) State evolution for $\dot{\theta}_0 = -0.095$ rad/s.



(d) Limit cycle for $\dot{\theta}_0 = -0.095$ rad/s.



(e) State evolution for $\dot{\theta}_0 = 1.8$ rad/s.



(f) Limit cycle for $\dot{\theta}_0 = 1.8$ rad/s.

Figure 11: State evolution and limit cycles of the rimless wheel for different initial angular velocities.

4.2 Effect of Simultaneous Parameter Changes

In this configuration, I modified all the key physical parameters: the leg length was reduced to $l = 0.7$ m, the inter-leg half-angle to $\alpha = \pi/10$, and the slope inclination increased to $\gamma = 0.12$ rad, while the initial angular velocity was kept fixed at $\dot{\theta}_0 =$

0.95 rad/s.
The time response (Figure 12) shows that the system rapidly converges to a periodic behavior. Both the angular position θ and the velocity $\dot{\theta}$ stabilize into oscillations with fixed amplitude and frequency. The phase portrait confirms this result: trajectories spiral inward toward a closed orbit, indicating convergence to a unique *limit cycle*.
Compared to the default case, (Fig. 11a) and (Fig. 11b), where the system settled to a static equilibrium and halted in double support, this new setup yields dynamic locomotion sustained by a stable oscillatory regime. The modified parameters effectively shift the equilibrium conditions: reducing l increases angular acceleration, reducing α sharpens inter-leg transitions, and increasing γ enhances gravitational drive. These combined effects allow the system to overcome energy dissipation at impacts and sustain walking motion.
Therefore, the modification results in a qualitatively different asymptotic behavior. The system now converges to a non-trivial limit cycle rather than a rest equilibrium. Only one such limit cycle is observed, suggesting a unique basin of attraction for the chosen initial condition.

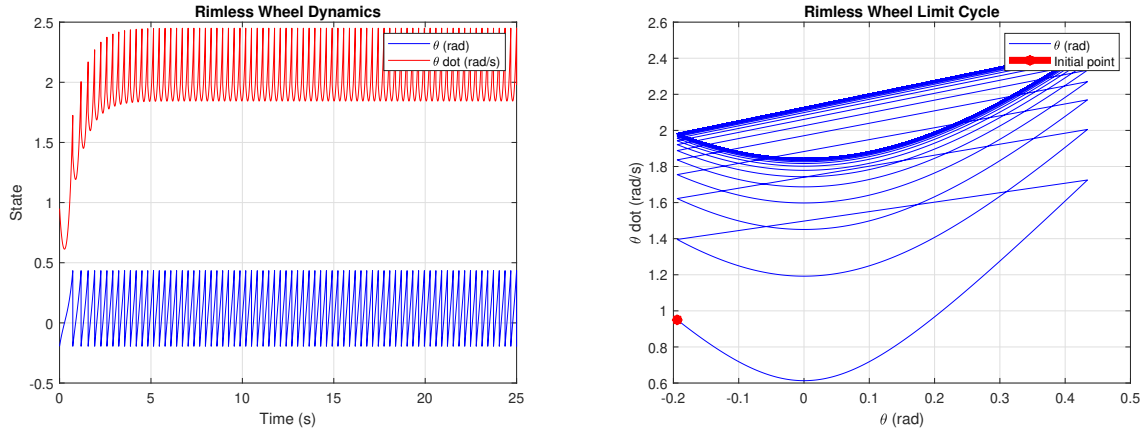


Figure 12: Performance comparison with changes of physical parameters.

References

[1] F. Ruggiero, *Lecture Slides – Underwater Robotics*, Field and Service Robotics, Università degli Studi di Napoli Federico II, 2025.

[2] F. Ruggiero, *Lecture Slides – Legged Robotics*, Field and Service Robotics, Università degli Studi di Napoli Federico II, 2025.

[3] OpenAI, *ChatGPT (June 2025 Version)*, `https://chat.openai.com`